Face moments from envelopes

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Abstract

The bundled moment hypotheses `FaceMoments` of the $d = 2$ cutoff theorems are discharged from polynomial-exponential envelopes of the Taylor data (Astra #5(c)). The key estimate is the finite-part bound

$$|\mathrm{FP}_\gamma(\Psi(\cdot, s))| \le \frac{b^{M-\gamma}}{M-\gamma} H(s) + \sum_{m<M} |f_m(s)|\,|\mathrm{axisPrim}(\gamma, b, m)|,$$

after which every face coefficient (monomial, finite part, logarithmic, and the remainder envelope) is bounded by a constant times $(1+s)^D e^{\beta s a'}$ and the Gaussian moment lemma applies; the exponents are automatically positive since the monomial exponent is $(m + h_1 + 1)/k_1$. The constructor `faceMoments_of_envelope` removes the opaque `FaceMoments` assumptions from applications. Zero `sorry`/`axiom`.

1 The things formalised

```
theorem axisFinitePart_abs_le (  b H :  ) (hb : 0 < b) (M :  ) (hM :   < M) (f :   →  )
    (fm :   →  ) (hrem :   v   Ioc (0 :  ) b, |taylorRem f fm M v|   H * v ^ M) :
    |axisFinitePart   b f fm M|
       H / ((M :  ) -  ) * b ^ ((M :  ) -  )
        +   m   Finset.range M, |fm m| * |axisPrim   b m|
noncomputable def faceScalar (  A :  ) (k  k  :  ) (m :  ) :
theorem faceCoeff_eq_scalar_mul ... :
    faceCoeff   A k  k  f m s = faceScalar   A k  k  m * f m s
theorem faceExp_pos (p  :  ) (h  k  :  ) (hk  : 0 < k ) (m :  ) :
    0 < faceExp p  ((k  :  ) * p  - h  - 1) k  m
theorem faceLogCoeff_abs_le ... :
    |faceLogCoeff   k  k  f M s|     m   Finset.range M, 1 / ((k  :  ) * k ) * |f m s|
theorem faceMoments_of_envelope (  p  b a' Cf CH :  ) (h  k  k  M D :  ) (h  : 0 <  )
    (hb : 0 < b) (hk  : 0 < k ) (hk  : 0 < k ) (hp  : 0 < p ) (Ψ :   →   →  )
    (hΨc : Continuous (Function.uncurry Ψ)) (f :   →   →  ) (hf :   m, Measurable (f m))
    (H :   →  ) (hH : Measurable H) (hH0 :   s, 0   H s) (h  : (k  :  ) * p  - h  - 1 < M)
    (hCH : 0   CH)
    (hfenv :   m, m < M →   s, 0 < s → |f m s|   Cf * (1 + s) ^ D * Real.exp (  * s * a'))
    (hHenv :   s, 0 < s → H s   CH * (1 + s) ^ D * Real.exp (  * s * a'))
    (hrem :   s, 0 < s →   u   Ioc (0 :  ) b,
      |Ψ u s -   m   Finset.range M, f m s * u ^ m|   H s * u ^ M) :
    FaceMoments   p  b h  k  k  M Ψ f H
```

2 Mathematics

The regularised integral is the tail estimate of unit 100 at $\varepsilon = b$: $|\int_0^b v^{-1-\gamma} R_M| \le H\, b^{M-\gamma}/(M-\gamma)$; the endpoint counterterms are bounded termwise. Each face coefficient is then dominated by a constant times the common envelope $(1+s)^D e^{\beta s a'}$: monomial coefficients are scalar multiples of f_m (`faceScalar`), the logarithmic coefficient is a sum of at most M such multiples, the finite part by the displayed bound, and the remainder envelope is a constant times H. The Gaussian moment lemma (`moment_integrableOn_of_envelope`) needs only the exponent $\alpha - 1 > -1$: the monomial exponents are $(m + h_1 + 1)/k_1 > 0$, the finite-part and log exponents are $p_2 > 0$, and the remainder exponent $p_2 + (M-\gamma)/k_1 > p_2$. Because the weight is Gaussian, any finite a' is admissible (Astra #5(c)).

3 Paper-facing meaning

With this constructor the cutoff theorems of units 107 and 109 need only: continuity of the data, compatible Taylor remainders of the face jets with envelopes $C(1+s)^D e^{\beta s a'}$, and the mixed remainder bound. For the chart amplitude $\eta e^{\beta s \xi}$ these are exactly the outputs of the smooth rectangular Taylor theory (Astra #5 ranks 3 and 5), the next units.

4 Lean notes

- `weighted_taylorRem_tail_le` at $\varepsilon = b$ gives the regularised-integral bound for free.
- Field goals generated by `refine ?_, ?_, ?_` for a `Prop` structure with `Fin M   Fin 2` indices: `rcases i with m | i` then `fin_cases i`, with `simpa [face , faceJ, faceC] using …` to align the unfolded index functions.
- A `set` variable and its unfolded value are defeq but not syntactically equal; goals created after the `set` (structure fields) show the unfolded form, and `rw` still closes them by `rfl`.
- `abs_of_nonneg (by positivity : (0 :  )   1 / (↑k  * ↑k ))` works without positivity hypotheses on the naturals.